

Trial Test 7

- Submit your answers via Google Forms (you have only 1 hour and 10 mins to complete).
- Your score will be available immediately after the test. Please, check your emails.
- If you have technical issues, let me know in DM or in the Group Chat.
- You are not allowed to use the calculator. Also, do not google, be honest!
- Do not panic and manage your time wisely. Read the questions carefully and try to answer all of them.

Good Luck!

1. Simplify:

$$(\sqrt{6} + \sqrt{3})^2 - (\sqrt{6} - \sqrt{3})^2$$

a) $12\sqrt{2}$

b) $6\sqrt{2}$

c) $4\sqrt{2}$

d) $10\sqrt{2}$

2. Simplify:

$$\frac{2x^2 + 3x + 1}{x^2 - 3x - 4}$$

a) $2 + \frac{7}{x-4}$

b) $2 + \frac{9}{x-4}$

c) $1 + \frac{5}{x-4}$

d) $2 + \frac{3}{x-4}$

3. Rearrange to give x in terms of y:

$$y = 3 + 2\left(1 - \frac{x}{3}\right)^2$$

a) $x = 3 - 3\sqrt{\frac{y-3}{2}}$

$$\text{b) } x = 3 + 3\sqrt{\frac{y-3}{2}}$$

$$\text{c) } x = 3 - 3\sqrt{\frac{y+3}{2}}$$

$$\text{d) } x = 3 + 3\sqrt{\frac{y+3}{2}}$$

4. Work out:

$$(2.1 * 10^8) * (6 * 10^{-5})$$

Write your answer in standard form:

$$\text{a) } 12.6 * 10^4$$

$$\text{b) } 1.26 * 10^3$$

$$\text{c) } 1.26 * 10^4$$

$$\text{d) } 12.6 * 10^3$$

5. The weight of a piece of wire is directly proportional to its length.

A piece of wire is 25 cm long and has a weight of 6 grams.

Another piece of the same wire is 30 cm long.

Calculate the weight of the 30 cm of wire.

$$\text{a) } 7.2$$

$$\text{b) } 6.4$$

$$\text{c) } 8.6$$

$$\text{d) } 3.2$$

6. The equation of a curve is $y = px^2 + qx$, where p and q are constants.

The curve passes through the points (2,6) and (4,-4).

What is the value of $q - p$?

a) -11

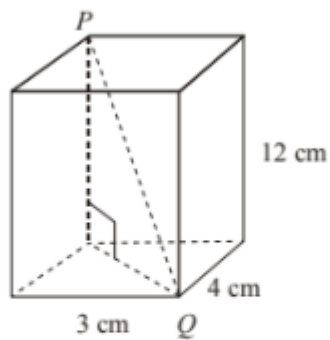
b) 9

c) 5

d) -2

7. A cuboid has length 3 cm, width 4 cm and height 12 cm.

Work out the length of PQ:



a) 16

b) 15

c) 11

d) 13

8. Express the recurring decimal $0.1(26)$ as a fraction:

a) $\frac{15}{198}$

b) $\frac{27}{198}$

c) $\frac{25}{198}$

d) $\frac{32}{198}$

9. The ball for a garden game is a solid sphere of volume 192 cm cubed.

For the children's version of the game the ball is a solid sphere made of the same material, but the radius is reduced by 25%.

What is the volume, in cm cubed, of the children's ball?

a) 96

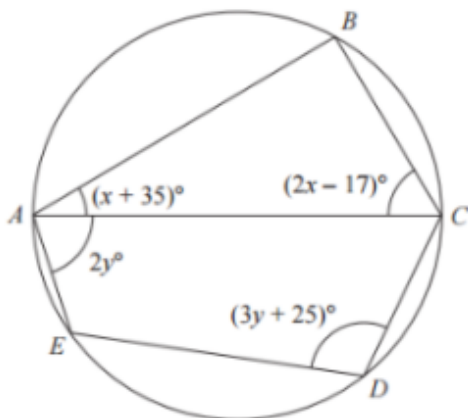
b) 81

c) 48

d) 108

10. ABCDE is a circle with diameter AC. Angle BAC = $x+3$ and angle BCA is $2x-17$.

What is the value of x?



a) 24

b) 27

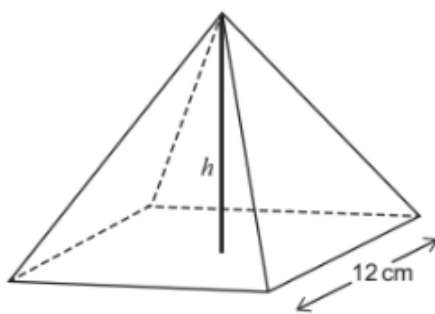
c) 16

d) 21

11. A solid pyramid has a square base of side length 12 cm and a vertical height of h cm.

The volume of the pyramid, in cm cubed, is equal to the total surface area of the pyramid, in cm cubed.

What is the value of h ?



[diagram not to scale]

- a) 16
- b) $4\sqrt{2}$
- c) 8
- d) $8\sqrt{2}$

12. Here is part of Gary's electricity bill.

Work out how much Gary has to pay for the units of electricity he used.

Electricity bill	
New reading	7155 units
Old reading	7095 units
Price per unit 15p	

- a) 1000p

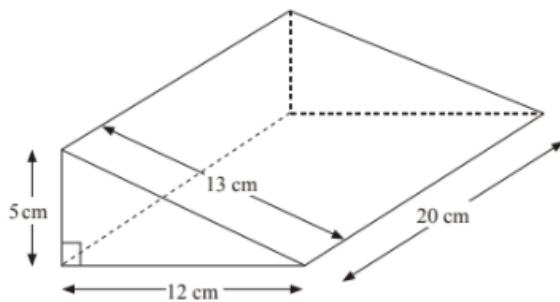
b) 900p

c) 850p

d) 750p

13. The diagram shows a right-angled triangular prism.

Work out its surface area.



a) 420

b) 380

c) 320

d) 400

14. For what values of x is $|16 - (x + 5)^2|$ at its minimum?

a) -1 and 4

b) -1 or -9

c) 1 and 9

d) -4 and -1

15. The symbol $\bullet$ defines a mathematical binary operation such that $a \bullet b = \sqrt{ab}$ for all positive integers.

Evaluate the $(8 \bullet 2) \bullet (4 \bullet 9)$

a) $\sqrt{32}$

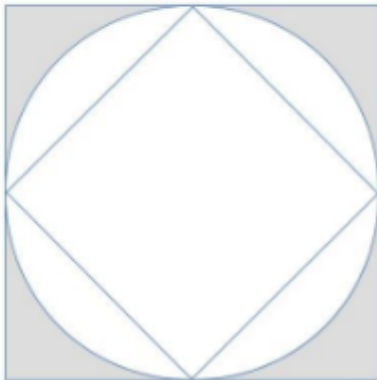
b) $\sqrt{24}$

c) $\sqrt{8}$

d) $\sqrt{20}$

16. The length of each side of the smaller square is $a\sqrt{2}$

What is the best expression for the shaded area?



a) $4a^2 - \pi a^2$

b) $4a^2 - \frac{\pi a^2}{2}$

c) $2a^2 - \frac{\pi a^2}{2}$

d) $2 - \pi a^2$

17. The shortest distance between two opposite sides of a regular hexagon is 12 cm.

Find its area.

a) $48\sqrt{3}$

b) $54\sqrt{3}$

c) $36\sqrt{3}$

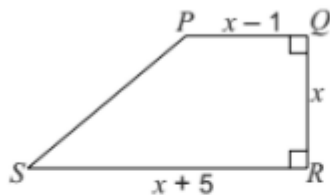
d) $72\sqrt{3}$

18. In trapezium PQRS, the parallel sides are PQ and RS.

$PQ = (x-1)$ cm, $RS = x+5$ cm and the vertical height $QR = x$ cm.

The area of the trapezium is 120 cm squared.

What is the length of RS?



a) 18

b) 10

c) 12

d) 15

19. A rhombus has diagonals of length 5 cm and 3 cm.

An enlargement of the rhombus has sides of length $\sqrt{68}$ cm.

What is the scale factor of the enlargement?

a) $2\sqrt{2}$

b) $2\sqrt{3}$

c) $4\sqrt{2}$

d) 8

20. Express 108 as the product of powers of its prime factors:

a) $2^2 * 3^3$

b) $2^3 * 3^2$

c) $2^3 * 3^3$

d) $2^2 * 3^2$

21. Simplify:

$$\left(\frac{27a^{-5}b^{-3}}{3^5a^{-7}b^{-5}} \right)^{-1}$$

a) $3^2 * a^{-2} * b^{-1}$

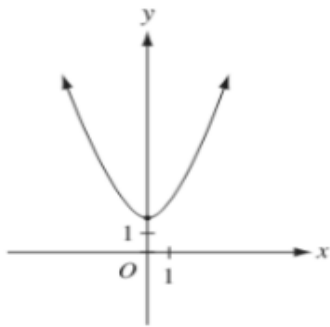
b) $3^2 * a^{-1} * b^{-2}$

c) $3^2 * a^{-1} * b^2$

d) $3^2 * a^{-2} * b^{-2}$

22. The graph of $y = x^2 + 2$ is shown below.

If the graph were to be shifted up 2 and left 3, which of the following would be the equation of the transformed function?



a) $y = (x - 3)^2 - 4$

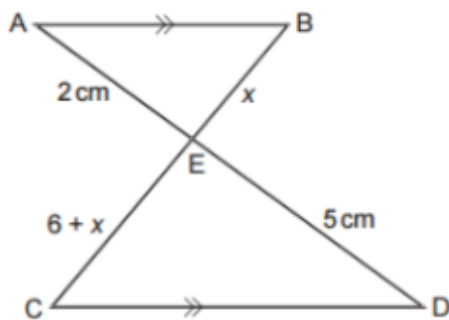
b) $y = (x + 3)^2 + 4$

c) $y = (x - 3)^2 + 4$

d) $y = (x + 3)^2 - 4$

23. AB is parallel to CD.

Find the length of BE.



a) 8

b) 12

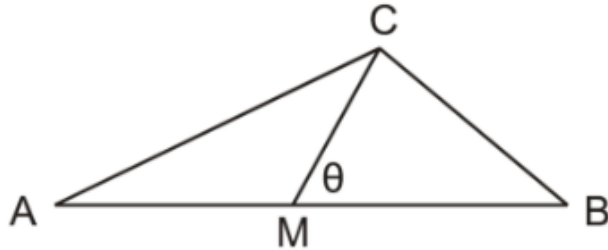
c) 10

d) 4

24. In the triangle ABC shown below

$\tan A = \frac{1}{6}$ and $\tan B = \frac{2}{3}$. M is the midpoint of AB.

What is the value of $\tan \theta$?



a) $\frac{7}{3}$

b) $\frac{5}{9}$

c) $\frac{2}{3}$

d) $\frac{4}{9}$

25. Find the positive solution of the following simultaneous equations:

$$4x^2 + y^2 + 10y = 47$$

$$2x - y = 5$$

a) $x=3, y=1$

b) $x=1, y=3$

c) $x=6, y=1$

d) $x=2, y=3$

26. Five bells commence tolling together and toll at intervals 3, 6, 9, 12, and 15 seconds respectively.

In 60 minutes, how many times do they toll together?

- a) 25
- b) 30
- c) 20
- d) 27

27. A product in a shop is reduced in price by 20%. At this reduced price the shopkeeper makes only 4% profit.

What percentage profit (to the nearest whole percent) does the shopkeeper makes at its normal selling price?

- a) 16
- b) 30
- c) 15
- d) 25

28. An exterior angle of a regular polygon with n sides is 4 degrees larger than an exterior angle of a regular polygon with $(n+3)$ sides.

What is the value of n ?

- a) 21
- b) 15
- c) 18
- d) 12

29. Buka has a collection of 5-tenge and 10-tenge coins worth a total of 400 tenge.

If he has 50 coins in all, how many 5-tenge coins does Buka have?

- a) 30
- b) 20
- c) 40
- d) 50

30. Imagine a plain, bog-standard clock that reads 5:40.

What is the angle formed between the hour and the minute hands?

- a) 105
- b) 100
- c) 70
- d) 75